PROJECT SUMMARY REPORT-OOP

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Project name: oop-Student Attandance Management

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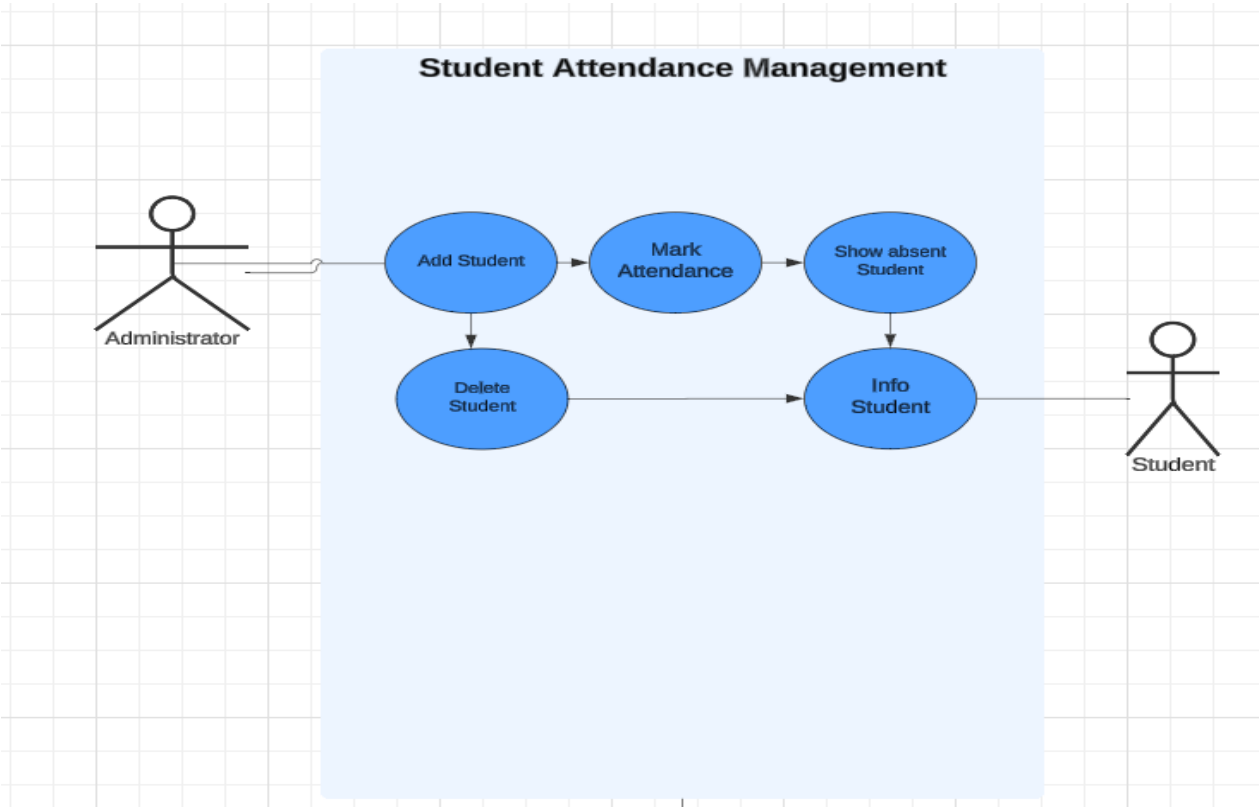
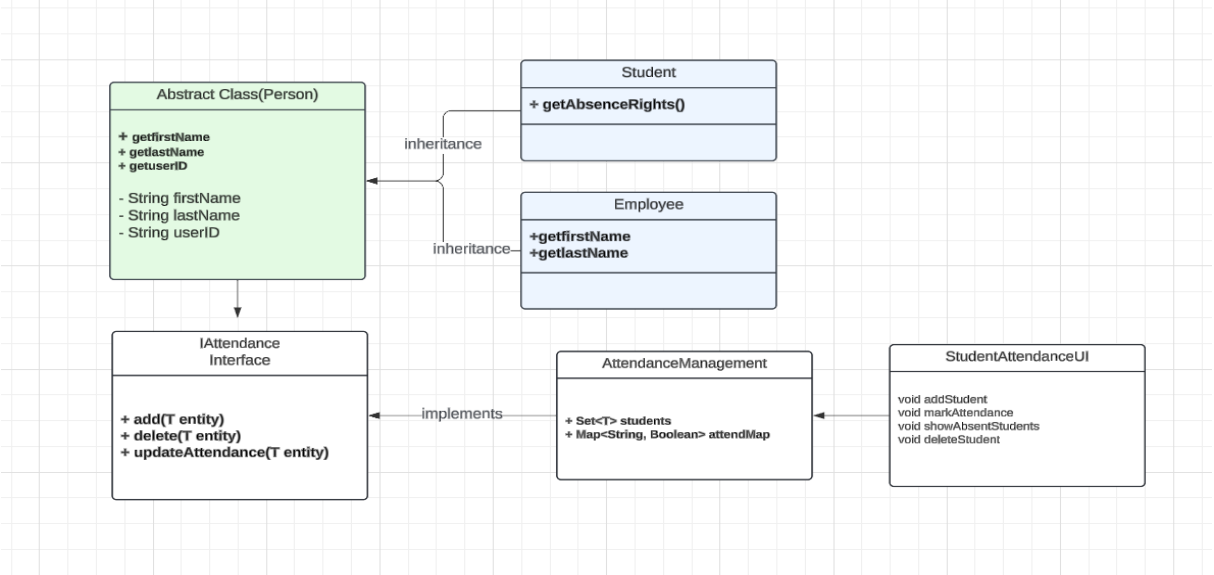
Business description of project topic:

Student Attandance Management project is a desktop-based application designed to take student attandance list based on their ıd that given by user. With help of UI Layer Design to create a window to user click on button and enter information take attandance list.

Drawing Use Case Diagram:



Drawing Class Diagram:

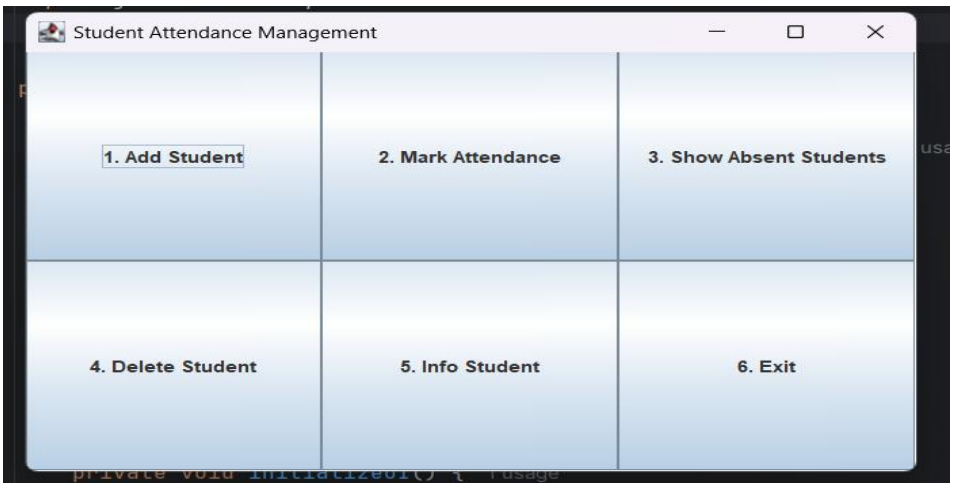


Explaining the design of overall system structure:

Student Attendance Management, the business layer is responsible for implementing the functions such as adding students, marking attendance, and tracking absence student and delete student and show student info.

Explaining UI Layer Design: Project's user interfaces:

UI Layer is Java Swing to create a graphical desktop application and we use this for our project. Accepting input from users for adding students, marking attendance delete students from the list.Also,show absent student and we did all of this with button use (JButton) and (JTextArea) to display information to the user etc. You can see our window in below.



When employee take attandance with click to mark attandence button it will show attandence list that marked student and when they click to Show Absent Student are going to appear in the JTextArea. When they click to add student button The user enters the firstname, lastname, and ID in text fields. Clicking the Add Student button sends the data to the business layer. When you click Delete Student button you need to enter student ID and after that there will be student deleted messages. If you click to exit the system is going to shut down.

Explaining Data Layer Design:

Data Layer Design is using Java collections. Such as, Set, Map, List to handle student records and managing attendance records. Set "HashSet" is used for storing student info such as, first name, last name, ID. Map "HashMap" is used for attendance status for each student by their ID with the help of Boolean false for absent code.

Software design details:

Explaining the implementation of

i. **Inheritance,**
ii. **Interface,**
iii. **Polymorphism,**
iv. **Generic class,**
v. **Generic collections (at least two of List, Set and Map),**
vi. **Lambda functions.**

Inheritance:

Establishes a parent-child relationship between classes.Base class is in this project is Person and Derived Class is Student and Employee. Studentand employee inherits the properties such as, firstName, lastName, userID and methods of Person.

Interface:

An interface is a contract that defines methods a class must implement. It supports abstraction and promotes polymorphism. IAttendance is our interface for attendance management.

Polymorphism:

Polymorphism has two side one of them is method overriding and other one is Interface-Based Polymorphism. The add and delete methods in AttendanceManagement override the IAttendance methods and Student java file has polymorphism due to super code that use person firstname etc.

Generic Class:

A generic class is a class that can work with different data types using type parameters.AttendanceManagement<T> is a generic class where T extends Person.

Generic Collection(List,Set and Map): Set used for store student info.Map used for tracking attendance and list is used for store Students all of info mostly for absent student.

Lambda Functions: Lambda functions are used in methods getAbsentStudents.